

Final Report: Freezer Optimization via Dynamic Programming

ECE 271C Project

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Problem Overview

This project considers the optimal operation of an industrial freezer over a 24-hour horizon under time-varying electricity pricing. Freezers possess significant thermal mass, allowing them to “store cold” by pre-cooling when electricity is cheap and coasting through expensive periods—without violating food-safety temperature constraints. Electricity prices vary up to fourfold across the day (\$0.05–\$0.20/kWh), and demand charges penalize peak power draw (\$18.50/kW, billed monthly). The goal is to design a cost-minimizing control strategy subject to temperature constraints, which is naturally formulated as a multi-stage decision problem under uncertainty and solved via dynamic programming.

Scenario 1: Finite Horizon Dynamic Programming

DP Formulation

State, Control, and Disturbance. The state at stage k is the current freezer temperature $T_k \in \mathbb{R}$. The control $P_k \in [0, P_{\max}]$ is the cooling power applied at that stage. The disturbance ω_k is a time-varying Gaussian random variable (using the hour-specific distribution table from the problem documentation rather than a single stationary distribution, to better emulate real operating conditions). The admissible control set enforces the temperature bounds: if T_k lies outside the safe band $[-25, -15]^\circ\text{C}$, the controller is forced to apply corrective action to return the state to the feasible range.

State Transition. The temperature evolves according to the freezer’s thermal dynamics:

$$T_{k+1} = T_k + \alpha(T_{\text{amb}} - T_k) - \beta P_k + \omega_k, \quad (1)$$

where α governs heat exchange between the freezer interior and the ambient environment, β is the compressor cooling efficiency, and ω_k is the stochastic disturbance.

Stage and Terminal Costs. The stage cost at hour k is

$$g_k(T_k, P_k) = c_k P_k \Delta t + \varphi_k(T_k), \quad (2)$$

where c_k is the time-varying electricity price, $\Delta t = 1$ hr, and $\varphi_k(T_k)$ is a quadratic penalty for violation of the safe temperature band $[-25, -15]^\circ\text{C}$. The terminal cost enforces a desired end-of-day temperature of -20°C :

$$g_N(T_N) = \varphi_N(T_N) = \begin{cases} 0 & \text{if } T_N = -20^\circ\text{C}, \\ \varphi_N(T_N) & \text{otherwise,} \end{cases} \quad (3)$$

where φ_N penalizes deviation from -20°C on a quadratic scale. Setting a specific terminal target ensures that the resulting policy is genuinely repeatable day-over-day (i.e., the freezer begins and ends each day at approximately the same temperature).

Objective. The objective is to minimize the expected total cost over the $N = 24$ stage horizon:

$$\min_{P_0, \dots, P_{N-1}} \mathbb{E} \left[\sum_{k=0}^{N-1} g_k(T_k, P_k) + g_N(T_N) \right]. \quad (4)$$

Uncertainty Model

Rather than using a single disturbance distribution, the simulation uses the time-varying Gaussian table provided in the problem documentation, sampled via `rng.gauss`. This more accurately reflects realistic conditions in which the magnitude and variance of thermal disturbances change throughout the day.

State and Action Space Discretization

The temperature grid is discretized in 0.1°C steps, padded 5°C beyond the safe band on each side, yielding a grid spanning $[-30, -10]^\circ\text{C}$. The power grid uses 0.05 kW steps from 0 to $P_{\max}$. At the temperature boundaries, the feasible action set collapses to a single override value, enforcing the boundary-return constraint.

Bellman backups are computed via antithetic sampling rather than direct integration over the disturbance distribution: the disturbance is sampled and the expectation is estimated from those samples, improving variance reduction. For Monte Carlo evaluation, 10,000 independent trials are run (with a fixed seed separate from the Bellman samples), each drawing 24 fresh Gaussian disturbances. Both the DP policy and a baseline thermostat policy using the same fixed-compressor freezer model are rolled out on identical disturbance paths for a fair comparison. Metrics collected per trial include total cost, temperature violations, final temperature deviation, and override activation counts.

The baseline thermostat is a full-on/full-off controller on that same compressor model: it applies full power if $T_k \geq -17^\circ\text{C}$, switches off if $T_k \leq -23^\circ\text{C}$, and otherwise holds the previous state.

Penalty Function Design

The non-terminal penalty $\varphi_k(T_k)$ applies a quadratic cost to violations of the safe temperature band $[-25, -15]^\circ\text{C}$, making larger violations proportionally more costly. The terminal penalty $\varphi_N(T_N)$ applies a quadratic cost to deviation from -20°C , with a weight significantly smaller than the non-terminal penalty—reflecting that ending at exactly -20°C is a periodicity convenience rather than a hard safety constraint.

Resulting Bellman Equation

The Bellman equation incorporates both the time-varying energy cost and the quadratic penalty terms, so the cost-to-go tracks not only the expected future energy bill but also the objective of keeping stored product within food-safe temperature limits. The backward recursion is:

$$V_k(T) = \varphi_k(T) + \min_{P \in \mathcal{U}(T)} \mathbb{E}_{\omega_k} [c_k P \Delta t + V_{k+1}(T + \alpha(T_{\text{amb}} - T) - \beta P + \omega_k)], \quad (5)$$

with $V_N(T) = g_N(T)$. Separating the penalty $\varphi_k(T)$ outside the minimization reflects that the temperature-violation cost at stage k is determined by the current state and is not a decision variable; only the energy cost and future cost-to-go are optimized over the choice of P .

What should the Out of Bands Costs be?

We really don't want the temperature of the freezer to leave the allowed temperature range. We have 2 options, an infinite cost, or a finite cost. Instead of an infinite cost, we choose to use a very large but finite cost. Below are the reasons why.

- A quadratic penalty is more accurate since the further the temperature is from the bounds, the more it is penalized. An infinite penalty treats them all the same way.
- An infinite penalty breaks the Bellman expectation. Each backup averages the interpolated future value over hundreds of disturbance samples — if even one sample lands out of bounds, that value is infinite, so the whole average (and thus the cost-to-go) becomes infinite.
- Having a penalty of, for example 10000 is high enough that the DP algorithm never selects it unless there is no choice (the compressor is not feasible for that location).
- Since the disturbance can theoretically be arbitrarily high, an infinite penalty would say "this outcome is infinitely bad and must never occur" whereas a finite penalty instead says "violations are very costly, so minimize their probability and severity," which is the real objective here.

What we had before:

Before, in the dynamic-programming backup, the expected cost-to-go $\mathbb{E}[V_{k+1}(T_{k+1})]$ was approximated using $N = 401$ fixed antithetic samples drawn directly from the disturbance distribution $w \sim \mathcal{N}(\mu_k, \sigma_k^2)$:

$$\mathbb{E}[V_{k+1}(T_{k+1})] \approx \frac{1}{N} \sum_{i=1}^N V_{k+1}(f(T_k, P_k) + w_i), \quad w_i \sim \mathcal{N}(\mu_k, \sigma_k^2).$$

However, when plotting the probability of failure vs. the out-of-bound penalty for Phoenix with Compressor C and $\alpha = 0.1$, the initial estimator produced the anomaly shown in Figure 1.

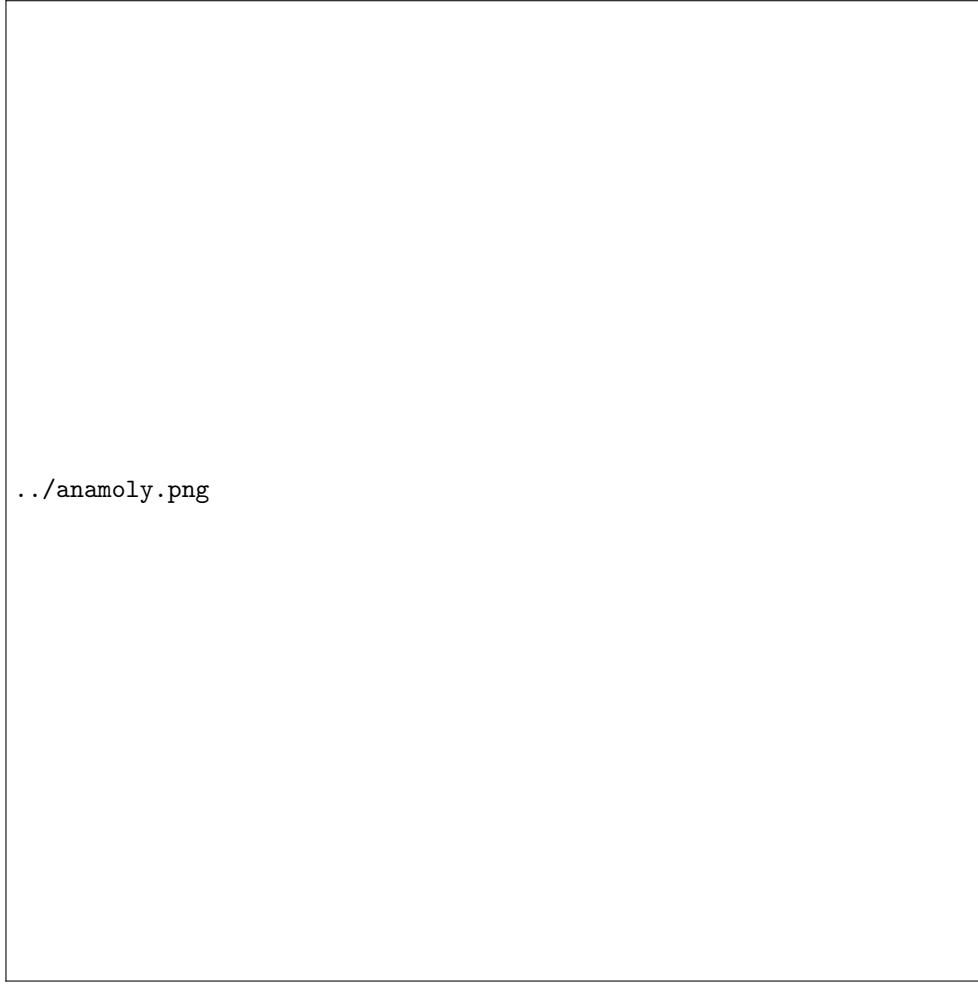


Figure 1: Anomaly in the original failure-probability sweep for Phoenix with Compressor C. The violation probability should decrease as the out-of-band penalty grows, but the naive disturbance estimator produced a non-monotone curve.

This is not supposed to happen. We would expect the probability of failure to keep decreasing as the penalty increases. The error in this was due to the fact that with only 401 draws, the sampled support reached only $\approx \pm 3\sigma$; rare, large disturbances were never represented. Because the out-of-band penalty is large, the value function is dominated precisely by those tail events, so the estimator systematically under-counted tail risk. This produced a suboptimal policy and the non-monotonic failure-vs-penalty curve.

What we do now (importance sampling):

We replaced the naive estimator with importance sampling, but still keeping the same number of samples. The $N = 401$ samples are now drawn from a widened proposal $q = \mathcal{N}(\mu_k, (c\sigma_k)^2)$ with $c = 3$, and each sample is reweighted by the likelihood ratio of the true density p to the proposal density q . This is to accurately show that these unlikely events sampled are more unlikely:

$$\mathbb{E}[V_{k+1}(T_{k+1})] \approx \sum_{i=1}^N \tilde{w}_i V_{k+1}(f(T_k, P_k) + x_i), \quad x_i \sim q, \quad \tilde{w}_i = \frac{p(x_i)/q(x_i)}{\sum_j p(x_j)/q(x_j)}.$$

The wider proposal places samples far into the tails, while the weights keep the estimator unbiased (tail samples receive correspondingly small weights). This lets the backup "see" the rare disturbances that drive constraint violations. The corrected Phoenix results are shown in Figure 2, and the corresponding violation probabilities and daily costs are tabulated in Figure 3.



Figure 2: Corrected failure-probability and energy-cost sweep for Phoenix with Compressor C after importance sampling is used in the Bellman expectation. Violation probability decreases sharply as the out-of-band penalty increases, while the daily energy cost changes only modestly.

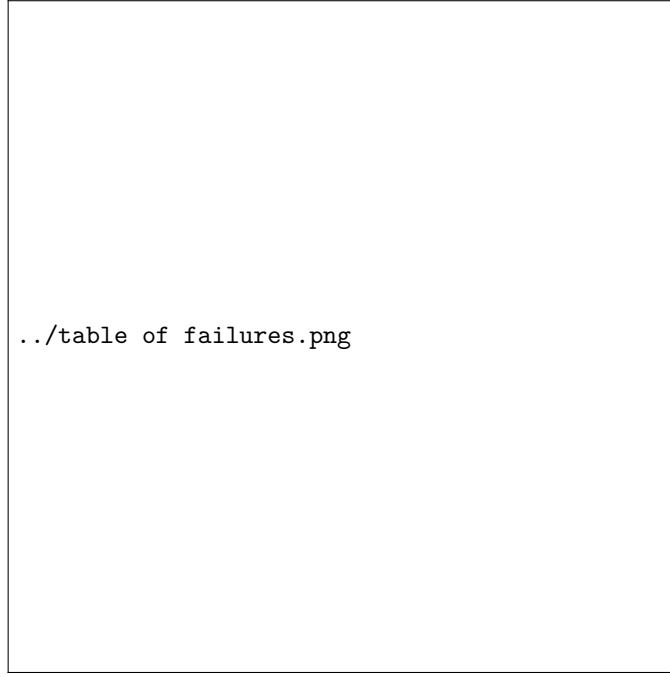


Figure 3: Numerical penalty sweep for Phoenix with Compressor C, showing the violation probability and mean daily energy cost at each out-of-band penalty weight.

Also, note that the only thing importance sampling changed in our results was the probability of failure. Everything else below for scenario 1 were nearly identical before this change.

Since $P(\text{violation})$ decreases substantially with the penalty till 100000 whereas the cost barely increased, we use the out of bounds penalty to be 100000 from here on out.

Scope note:

Importance sampling is applied only to the expectation inside the DP solve. The out-of-sample validation continues to draw disturbances from the true time-varying distribution $\mathcal{N}(\mu_k, \sigma_k^2)$ via `rng.gauss`, so the reported failure probabilities remain unbiased estimates of true performance.

We make the terminal penalty’s weight 100, significantly less, since it is more “nice to have” to end at -20°C than a hard requirement. It is a periodicity knob, not a safety one.

How does Alpha affect things?

We plot α against cost and $P(\text{failure})$ in Figure 4. Note that α is a property of the freezer, representing its thermal leakage.

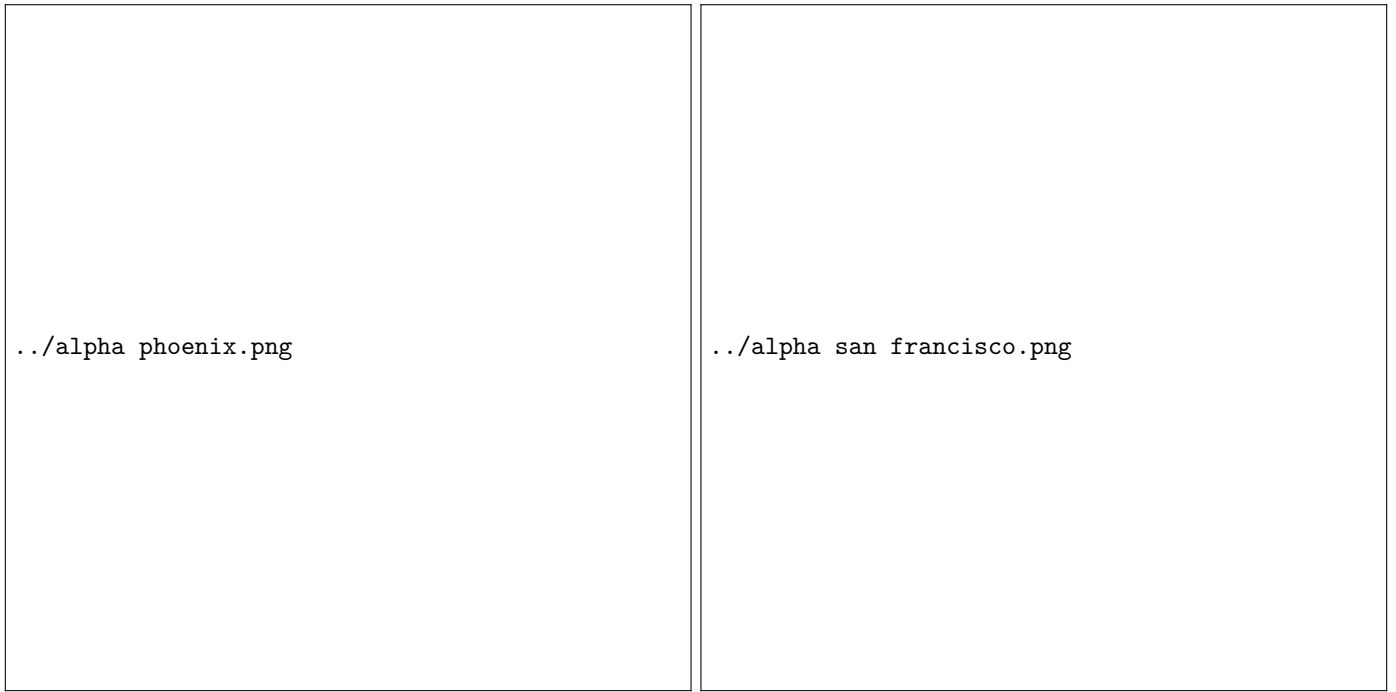


Figure 4: Effect of thermal leakage α on daily energy cost and violation probability for Phoenix and San Francisco with Compressor C. Higher α increases the heat load; in Phoenix, sufficiently high leakage also becomes a feasibility problem.

- The cost increases with α .
- In cold and mild climates, insulation (α) is a cost lever: it changes the bill but barely affects safety.
- In a hot climate, insulation becomes a feasibility lever: bad enough insulation makes the freezer literally uncoolable.
- We will use $\alpha = 0.1$ as a good medium point.

Which compressor to use?

Compressor C is the only feasible compressor at $\alpha = 0.1$ for all locations, as shown in Figure 5.

```
../compressor phoenix.png
```

```
../compressor SF.png
```

If α was lower, in colder climates, A and B might be competitive since they have lower capital costs, but C still wins on a lower energy bill due to its lower β .

How Necessary is Stochastic DP? If we used Deterministic DP with the disturbance just equal to its mean (no standard deviation), below is what happens. (Phoenix, SF, Minneapolis resp.)


```
../stoch vs det phoenix.png
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```
../stoch vs det SF.png
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If it is deterministic, the optimizer would have no reason to keep any safety margin. The cheapest deterministic policy rides the temperature right up against the band edges: it will happily sit at exactly -15°C through the expensive hours because, with no uncertainty, hugging the boundary is free.

Interestingly, the cost is barely lower in deterministic DP than stochastic. This is because all we are doing is cooling a touch less by hugging the edge instead of staying a degree or two below it.

Compared to the Thermostat In Phoenix (and the others), the cost reduction gained by DP is not much (both usually apply lots of power or very little power).



Figure 7: Mean daily energy-cost comparison between the stochastic DP policy and the thermostat baseline. The DP policy lowers cost modestly, but the larger benefit comes from reducing temperature violations.

The real benefit of using DP is the orders of magnitude difference in the probability of violation, illustrated in Figure 8. The thermostat makes the temperature violate the range 3 times. The reason for this is it does not react until the temperature has already crossed -17°C , and in Phoenix’s heat it overshoots past -15°C before it can pull back. On the other hand, the DP algorithm anticipates this.



Figure 8: Temperature and compressor-power trajectories for the DP policy and thermostat baseline. The DP policy anticipates future heat load and price periods, while the thermostat reacts only after its switching thresholds are reached.

Scenario 2: Demand Charges and Geographic Variation

Motivation and Difference from Scenario 1

Scenario 2 extends the one-day freezer scheduling model by introducing a demand charge and by evaluating the controller across three climate zones. In Scenario 1, the only direct operating cost was hourly energy consumption, so the DP mainly decided when to buy electricity: pre-cool during low-price hours and coast during high-price hours. Real utility bills often include a demand charge, where the customer is billed according to the single highest power draw over the billing period. This changes the problem because one full-power hour can dominate the demand-charge portion of the bill.

Therefore, the strategy can no longer only answer “when is energy cheap?” It must also shape the whole power profile and avoid creating unnecessary peaks. We also move from a Phoenix-only ambient profile to three locations: Phoenix, San Francisco, and Minneapolis. These represent hot desert, temperate coastal, and cold northern operating conditions, allowing us to evaluate how the policy adapts when the ambient temperature load on the freezer changes.

The reason this remains a DP problem is that the problem is still sequential, stochastic, and state-compressible. Every hour, the controller chooses a compressor power that affects next-hour temperature. The disturbance remains Gaussian and time-varying, so the Bellman recursion must take an expectation over random thermal shocks. The new issue is that peak power is not stage-decomposable. To preserve the Markov property, we augment the state with the peak power so far.

Component	Scenario 1	Scenario 2
Cost	Time-of-use energy cost	Energy cost plus amortized demand charge
State	Current temperature T_k	Temperature and peak power (T_k, M_k)
Location	Phoenix only	Phoenix, San Francisco, Minneapolis
DP structure	One-dimensional DP	Two-dimensional DP
Tradeoff	Pre-cool vs. coast	Pre-cool, coast, and avoid new peak demand

State, Control, Disturbance, and Transition

The planning horizon remains one day with $N = 24$ hourly stages and $\Delta t = 1$ hr. The Scenario 2 state is

$$x_k = (T_k, M_k), \quad (6)$$

where T_k is freezer temperature and

$$M_k = \max_{0 \leq j < k} P_j \quad (7)$$

is the highest compressor power used before the current hour. The initial state is $T_0 = -20^\circ\text{C}$ and $M_0 = 0$.

The control is compressor power,

$$u_k = P_k, \quad 0 \leq P_k \leq P_{\max}. \quad (8)$$

In the implementation, $P_{\max} = 12\text{ kW}$ and the control grid is $\{0, 0.5, 1.0, \dots, 12.0\}$ kW. The disturbance ω_k is the same time-varying Gaussian thermal disturbance used in Scenario 1:

$$\omega_k \sim \mathcal{N}(\mu_k, \sigma_k^2), \quad (9)$$

with hour-specific means and variances from the project handout.

The admissible control set is state-dependent: at the temperature boundaries it collapses to a single override value, and inside the safe band it is the full power grid,

$$\mathcal{U}_k(T_k, M_k) = \begin{cases} \{0\} & T_k < T_{\min}, \\ \{P_{\max}\} & T_k > T_{\max}, \\ \{0, 0.5, \dots, P_{\max}\} & T_{\min} \leq T_k \leq T_{\max}. \end{cases} \quad (10)$$

The first two cases encode the safety recovery rule: if the freezer drifts below $T_{\min}$ the compressor is forced off, and if it drifts above $T_{\max}$ it is forced to full power, returning the state toward the feasible band. Inside the band the DP policy chooses freely from the normal power grid. Because this admissible set is used both in the Bellman backup and in simulation, the boundary-return behavior is part of the policy itself rather than a separate rule imposed after the fact.

The transition has two pieces. Temperature evolves according to the same physical model as Scenario 1,

$$T_{k+1} = T_k + \alpha(T_k^{\text{amb}, \ell} - T_k)\Delta t - \beta P_k \Delta t + \omega_k, \quad (11)$$

where the location index ℓ selects the ambient temperature profile for Phoenix, San Francisco, or Minneapolis. The peak-power state evolves deterministically:

$$M_{k+1} = \max(M_k, P_k). \quad (12)$$

Thus the full state transition is

$$x_{k+1} = f_k(x_k, u_k, \omega_k) = \left(T_k + \alpha(T_k^{\text{amb}, \ell} - T_k)\Delta t - \beta P_k \Delta t + \omega_k, \max(M_k, P_k) \right). \quad (13)$$

How the Demand Charge Enters the DP

The key modeling question in Scenario 2 is how to introduce the demand charge into the Bellman recursion. The answer is that the demand charge does not enter the ordinary stage cost directly; instead, it appears in the terminal cost. The reason is structural. A stage cost must be a function of the current state, current control, and current disturbance. The demand charge depends on

$$\max_{0 \leq k < N} P_k, \quad (14)$$

which is defined over the whole horizon and cannot be evaluated from a single hour alone. This is exactly why M_k is added to the state. Once the state includes peak power so far, the terminal state contains the final peak M_N , and the demand charge can be evaluated without breaking Bellman's principle.

Define the soft safety penalty

$$\phi(T) = [\max(0, T - T_{\max})]^2 + [\max(0, T_{\min} - T)]^2. \quad (15)$$

The stage cost is

$$g_k(x_k, u_k, \omega_k) = c_k P_k \Delta t + \lambda_v \phi(T_k). \quad (16)$$

This stage cost contains only the immediate time-of-use energy spend and the soft safety penalty. The weight λ_v is a tunable violation penalty that controls the tradeoff between lowering electricity cost and reducing the probability that temperature exits the safe band. The demand charge itself is absent from the stage cost.

Instead, the demand charge appears in the terminal cost:

$$g_N(x_N) = \lambda_v \phi(T_N) + \lambda_f [\max(0, T_N - T_0)]^2 + \frac{c_d}{30} M_N. \quad (17)$$

This terminal cost has three pieces. The first keeps the safety penalty active at the final state. The second is a one-sided quadratic terminal pull-back toward $T_0 = -20^\circ\text{C}$; without this term, the optimizer could exploit the end of the horizon by shutting the compressor off in the last hour and letting the temperature drift upward. The third term, $(c_d/30)M_N$, is the daily-amortized demand charge.

The Bellman recursion is therefore

$$V_k(T, M) = \min_{P \in \mathcal{U}_k(T, M)} \left\{ c_k P \Delta t + \lambda_v \phi(T) + \mathbb{E}_{\omega_k} [V_{k+1}(T + \alpha(T_k^{\text{amb}, \ell} - T)\Delta t - \beta P \Delta t + \omega_k, \max(M, P))] \right\}, \quad (18)$$

with terminal condition given by $g_N(x_N)$.

Objective and Interpretation

For a policy μ , the objective is

$$J_\mu = \mathbb{E} \left[\sum_{k=0}^{N-1} g_k(x_k, \mu_k(x_k), \omega_k) + g_N(x_N) \right]. \quad (19)$$

Expanding the terms, the first part is the expected time-of-use energy cost C_{energy} . The second is the daily-amortized demand charge $C_{\text{demand}}/30$. The remaining terms are soft safety and terminal penalties. These penalty terms are not part of the utility bill; they are guidance terms that handle the unbounded Gaussian disturbance tails and keep the computed policy physically sensible. Therefore, the DP target captures the operator's bill while adding the penalty structure needed to maintain reliable freezer operation.

Thirty-Day Demand Charge Assumption

The project handout states that the demand charge is calculated over a 30-day billing period. It also states that, for simplicity, the 24-hour optimal control strategy may be assumed to repeat each day of the month. Under this repeatable-day assumption, the maximum power in the representative day is the same maximum power that would appear each day. Therefore, the monthly demand charge

$$c_d \max_k P_k \quad (20)$$

is converted into an equivalent daily charge:

$$\frac{c_d}{30} \max_k P_k = \frac{18.50}{30} M_N. \quad (21)$$

Thus, dividing by 30 is consistent with the handout and allows energy cost and demand cost to be reported on the same daily basis.

This does not mean that a real month must have identical temperatures every day. It is a simplifying assumption made because the handout provides representative 24-hour temperature profiles rather than 30 distinct daily profiles. If actual month-long temperature data were available, they could be fed into a multi-day or rolling-horizon DP. The state would then carry the month-to-date peak

$$M_t = \max_{0 \leq i < t} P_i \quad (22)$$

over the full billing period. The controller could then adapt to different daily temperatures, using less power on cooler days and reserving higher peak power for hotter or higher-disturbance days.

State-Space Discretization and Computational Cost

Augmenting the state with the peak power is the right modeling move, but it is not free: it turns Scenario 1’s one-dimensional temperature grid into a two-dimensional grid, $|\mathcal{X}| = |T\text{-grid}| \times |M\text{-grid}|$. Since one Bellman backup scales as states $\times$ controls $\times$ disturbance samples, the extra dimension multiplies an already large number.

Table 1 shows the blow-up. Reusing Scenario 1’s 0.05 kW resolution for the peak dimension ($\Delta M = 0.05$ kW) pushes the state count past 48,000 and the per-stage cost to $\approx 4.7 \times 10^9$, roughly 250 $\times$ Scenario 1 and too slow to solve across all 24 stages within our budget.

We recover tractability by coarsening the power step to $\Delta P = 0.5$ kW (the control grid already used above), tying the peak grid to it ($\Delta M = \Delta P$), and halving the disturbance samples to $N_s = 201$. The leverage is that ΔP does double duty: it sets the control resolution and, through ΔM , the peak-state resolution, so coarsening it shrinks both at once. With the sample cut on top, that is the $\approx 185\times$ saving in Table 1, and the per-stage cost falls back to $\approx 2.5 \times 10^7$, the same order as Scenario 1.

Table 1: Per-stage DP backup cost (states $\times$ admissible controls $\times$ disturbance samples) as the model moves from the one-dimensional Scenario 1 to the augmented two-dimensional Scenario 2. Tying $\Delta M = \Delta P$ and coarsening to 0.5 kW returns the per-stage cost to the Scenario 1 order.

Configuration	Controls		States	Samples	Per-stage cost
S1 ($\Delta P = 0.05$)	241		201	401	$\sim 1.9 \times 10^7$
S2 naive ($\Delta M = \Delta P = 0.05$)	241	$201 \times 241 = 48,441$		401	$\sim 4.7 \times 10^9$
S2 adopted ($\Delta M = \Delta P = 0.5$, $N_s = 201$)	25	$201 \times 25 = 5,025$		201	$\sim 2.5 \times 10^7$

The choice $\Delta M = \Delta P$ is also what keeps the coarsening close to lossless on the dimension that matters. With the peak and the control on one grid, $M_{k+1} = \max(M_k, P_k)$ always lands on an exact node, so the demand charge $(c_d/30)M_N$ is tracked without rounding. All we give up is control granularity (0.5 kW steps instead of 0.05), a cheap trade for a policy whose savings come from coarse pre-cool-and-coast shaping rather than fine power trimming.

Scenario 2: Results and Interpretation

Evaluation Protocol

We solve the two-dimensional Bellman recursion once per location and then test the resulting policy out of sample with a Monte Carlo rollout of 5,000 independent trials. A trial draws 24 fresh hourly disturbances $\omega_k \sim \mathcal{N}(\mu_k, \sigma_k^2)$ from the project’s time-varying table, and the DP policy and the fixed-compressor Scenario 1 thermostat baseline are run on the *same* disturbance path so the comparison is paired. For every trial we record the energy cost, the amortized demand charge $(c_d/30)M_N$, the realized peak power, and whether the temperature ever left the safe band $[-25, -15]^\circ\text{C}$. The comparison policy is the same full-on/full-off thermostat baseline defined in Scenario 1.

Cost and Reliability Across the Three Climates

Table 2 collects the headline numbers, and Figure 9 visualizes the cost split. The pattern is consistent across all three locations: the DP policy is cheaper than the fixed-compressor thermostat baseline on every cost component and, at the same time, keeps the freezer inside the safe band on roughly 97% of trials, while that thermostat baseline violates the band on essentially every trial. The thermostat’s violations are not marginal. Its 95th-percentile maximum temperature sits near -10 to -11°C , several degrees above the -15°C ceiling, whereas the DP’s 95th-percentile maximum barely touches the boundary at about -15.2°C . The thermostat is a reactive full-on/full-off controller with no notion of price or of the day ahead, so it lets the box drift toward its warm setpoint and gets caught there when the ambient load spikes. The DP instead pre-cools while electricity is cheap and banks a thermal buffer.

Table 2: Monte Carlo results by location (5,000 trials, seed 20260519). Costs are daily USD; the demand charge is the monthly \$18.50/kW amortized over 30 days. “Viol. rate” is the fraction of trials with at least one hour outside $[-25, -15]^{\circ}\text{C}$.

Location	Controller	Energy	Demand	Total	5–95% Total	Viol. rate
Phoenix	DP	\$27.32	\$7.07	\$34.39	\$33.11–35.70	3.06%
Phoenix	Thermostat baseline	\$28.16	\$7.40	\$35.56	\$33.20–37.40	100%
San Francisco	DP	\$19.51	\$5.36	\$24.87	\$23.46–26.73	2.84%
San Francisco	Thermostat baseline	\$21.35	\$7.40	\$28.75	\$26.60–30.80	99.9%
Minneapolis	DP	\$16.98	\$5.05	\$22.03	\$20.63–24.01	3.28%
Minneapolis	Thermostat baseline	\$18.39	\$7.40	\$25.79	\$24.20–27.20	100%

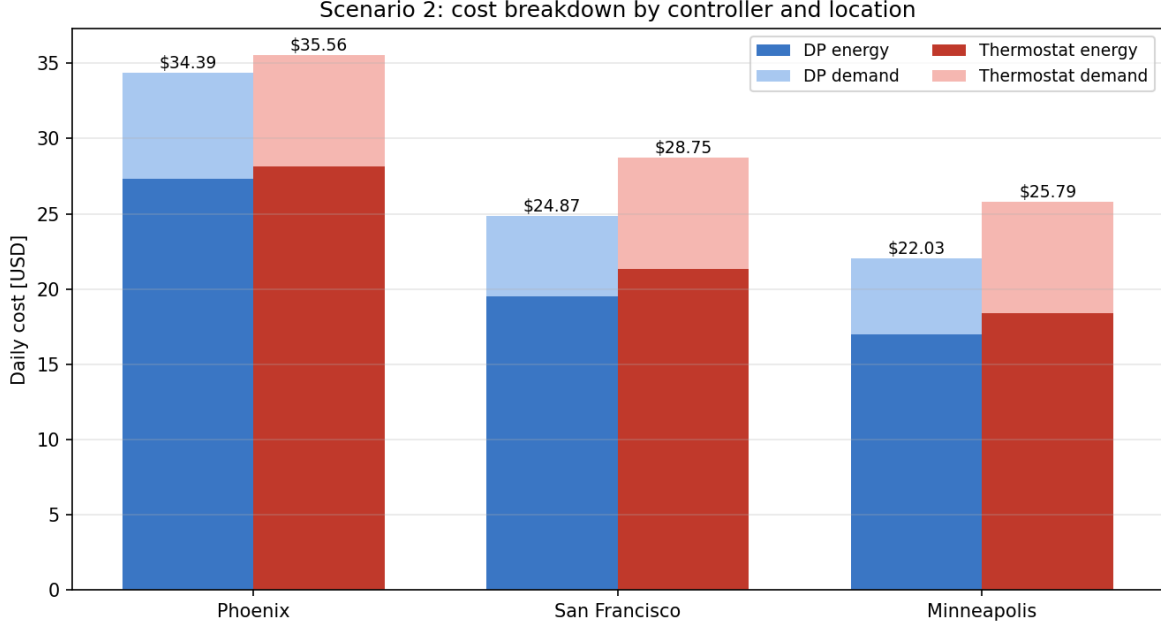


Figure 9: Daily-cost breakdown (energy plus amortized demand charge) for the DP policy and the fixed-compressor thermostat baseline in each climate. The DP is lower on both components in all three locations; its separate reliability advantage—holding the safe band on roughly 97% of trials against the baseline’s near-total violation—is reported in Table 2.

The spread columns show how far apart the two controllers really are. In Minneapolis the DP’s 95th-percentile day (\$24.01) lands below the fixed-compressor thermostat baseline’s 5th-percentile day (\$24.20), so the two cost distributions are nearly disjoint: an unusually expensive DP day still beats an unusually cheap thermostat-baseline day. San Francisco comes within a hair of the same separation, its two distributions just touching at \$26.6–26.7. Phoenix is the one place where they clearly overlap, the first sign that the hot-desert case is qualitatively different from the other two.

Structure of the Optimal Policy

Under the mean disturbance the DP temperature and power profiles trace a clear pattern (Figure 10). During the early cheap hours (\$0.05/kWh) the compressor runs hard and drives the temperature down toward the cold edge of the band, reaching roughly -24°C around hours 6–8. That stored cold is then spent through the expensive midday block (\$0.20/kWh), where the compressor backs off and the temperature is allowed to drift upward, climbing back toward -19 to -22°C without crossing -15°C . A second, smaller pre-cool appears in the evening before the day ends near the -20°C target. This is exactly the “pre-cool when cheap, coast when expensive” behavior we expected, and the terminal pull-back toward T_0 is what keeps the policy repeatable day to day (the DP’s terminal-violation rate is 0%, against 36–74% for the fixed-compressor thermostat baseline).

What changes across locations is the power level, not the strategy. Phoenix faces a mean ambient near 31°C , so heat leaks in fast and the compressor has to run near 11 kW for most of the day just to hold the line. Minneapolis sits near 10°C ambient, the heat load is light, and the same qualitative policy is achievable with peaks around 7.5–8 kW. San Francisco, at roughly 16°C , falls in between. The freezer physics are identical in all three runs; the climate sets how hard the policy

has to push.

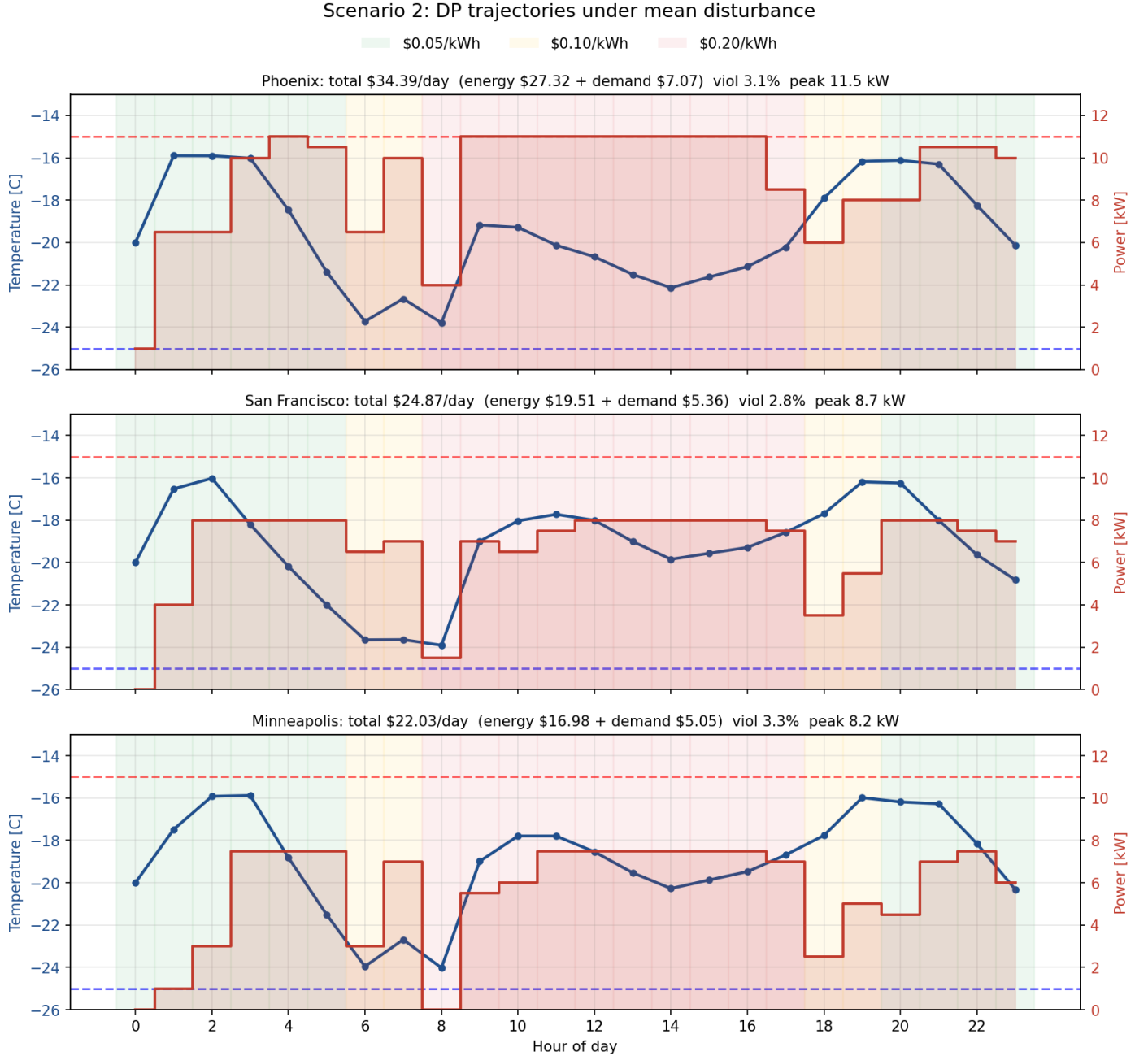


Figure 10: DP temperature (blue, left axis) and compressor power (red step, right axis) under the mean disturbance for each climate, with the time-of-use price blocks shaded. The policy pre-cools toward the cold edge of the band during the cheap early hours and coasts upward through the expensive midday block in all three cities; only the power level scales with the ambient load (Phoenix near 11.5 kW, Minneapolis near 8 kW). The dashed lines mark the safe band $[-25, -15]$ °C.

Where the Savings Come From: Energy Timing Versus Peak Shaping

The demand charge is the new ingredient in Scenario 2, so beyond the size of the saving there is a sharper question: which part of the bill does the DP attack? Table 3 splits each location's total saving into an energy piece and a demand piece. Read top to bottom, the demand share of the saving climbs from 28% in Phoenix to 62% in Minneapolis, and the overall saving grows from a thin 3.3% to a solid 14.6%. The mechanism behind the rotation is the peak-power constraint imposed by climate. Across the 5,000 trials the thermostat baseline pins its peak at 12 kW in every trial and every location, because it answers a warm box with full power. The DP's peak is instead a spread, and where that spread lands depends on the heat load. In Phoenix the load is high enough that the DP cannot drop far below the cap; its peak averages about 11.5 kW, so it has little room to undercut the thermostat baseline's \$7.40 demand charge and saves only \$0.33 there. In

Table 3: Savings decomposition, DP relative to the fixed-compressor thermostat baseline. The energy/demand split of the saving rotates with climate even though the policy logic is unchanged.

Location	Energy save	Demand save	Total save	Energy share	Demand share	vs. thermostat baseline
Phoenix	\$0.84	\$0.33	\$1.17	71.5%	28.5%	3.3%
San Francisco	\$1.84	\$2.04	\$3.88	47.4%	52.6%	13.5%
Minneapolis	\$1.42	\$2.35	\$3.77	37.6%	62.4%	14.6%

Minneapolis the load is light, the DP holds its peak near 8 kW, and the demand saving balloons to \$2.35. The demand charge rewards a flatter power profile, and only the mild climates give the controller enough slack to deliver one.

There is a subtler point hiding in the energy column. The DP is not winning on energy by using less electricity. In Phoenix it actually draws *more* total energy than the fixed-compressor thermostat baseline (215 kWh against 209 kWh) yet still pays about \$0.84 less for it, purely because it buys that energy during the cheap hours and avoids the expensive block. In San Francisco the two controllers consume almost the same 158 kWh, and the DP’s \$1.84 energy saving is again entirely a matter of *when*. Only in Minneapolis does the DP also use modestly less energy. The takeaway is that the energy saving is a load-shifting result, not a consumption-reduction result, which is the right behavior for a time-of-use tariff.

Robustness and Parameter Sensitivity

Because every number above is a stochastic rollout, we re-ran each location with five independent seeds (5,000 trials each) to rule out a lucky draw. The mean total cost shifts by at most \$0.05 and the violation rate by at most 0.54 pp (Phoenix \$34.39–\$34.42, San Francisco \$24.87–\$24.91, Minneapolis \$22.03–\$22.07). The policy is solved once and reused, so this is pure evaluation noise, small against the \$1–4 savings.

Two parameters set the operating point. The demand-charge rate c_d controls how hard the policy flattens its profile, and the climate split in Table 3 is itself a c_d sensitivity result: the same c_d yields very different demand savings depending on the thermal headroom. The violation weight λ_v sets the residual $\sim 3\%$ violation rate; that the high- and low-side counts are roughly equal (1.5% each in Phoenix) and the override fires on only $\sim 1.5\%$ of trials points to rare Gaussian tails, not a systematic bias. Raising λ_v lowers the residual rate at the cost of extra insurance cooling. A full sweep over c_d and λ_v is the natural next experiment.

Conclusion

Augmenting the state with the running peak power M_k is enough to fold a demand charge into a finite-horizon DP without breaking Bellman’s principle: M_k keeps the problem Markov, and the charge enters through the terminal cost rather than any stage cost. The resulting two-dimensional policy beats the fixed-compressor thermostat baseline in all three climates at once, cutting the daily bill by \$1.17 in Phoenix and up to \$3.88 in San Francisco while holding temperature inside the safe band on about 97% of trials, against a thermostat baseline that leaves it on almost every day.

The decomposition carries the main practical lesson. A demand-aware controller is not equally worth deploying everywhere: the same policy logic returns only a 3.3% saving in hot, load-bound Phoenix but 14.6% in mild Minneapolis, and the demand charge supplies a growing share of that gain as the ambient load falls. For a freezer in a temperate or cold climate, the payback comes mostly from reshaping the power profile rather than from buying less energy.

The main caveat is the repeatable-day assumption behind $(c_d/30)M_N$: a real billing month is set by its single worst day, not by an average one, so our daily numbers are a representative-day estimate rather than a true monthly bill. The cleanest fix is the rolling-horizon formulation already sketched above, where the state carries the month-to-date peak so the controller can spend a high peak only on genuinely hot days. Beyond that, sweeping c_d and λ_v , replacing the soft safety penalty with a chance constraint, and testing under a disturbance model the policy was not trained on would round out the picture.

Parameter Testing

Until now, we have introduced lots of variables in our algorithm that can be adjusted, such as discretization level, penalty parameters, and number of sample disturbances taken. In this section, we test how adjusting these variables affects the performance of our algorithm.

Penalty Parameter

Figures 1, 2, and 3 are heatmaps that show how changing the penalty parameter and terminal penalty parameter λ_v . As a reminder, these are used to scale the punishment for falling out of range of $T_{\min}$ and $T_{\max}$. The only difference between these two is the terminal penalty parameter is the penalty applied at the end of the day (AKA the last time step of our DP). We decided to independently measure these two because we believed there would be cases where we would want a higher terminal penalty parameter. However based on our results, this does not seem necessary.

For each location, there are two heatmaps, one for measuring the expected daily cost of applying our DP policy and

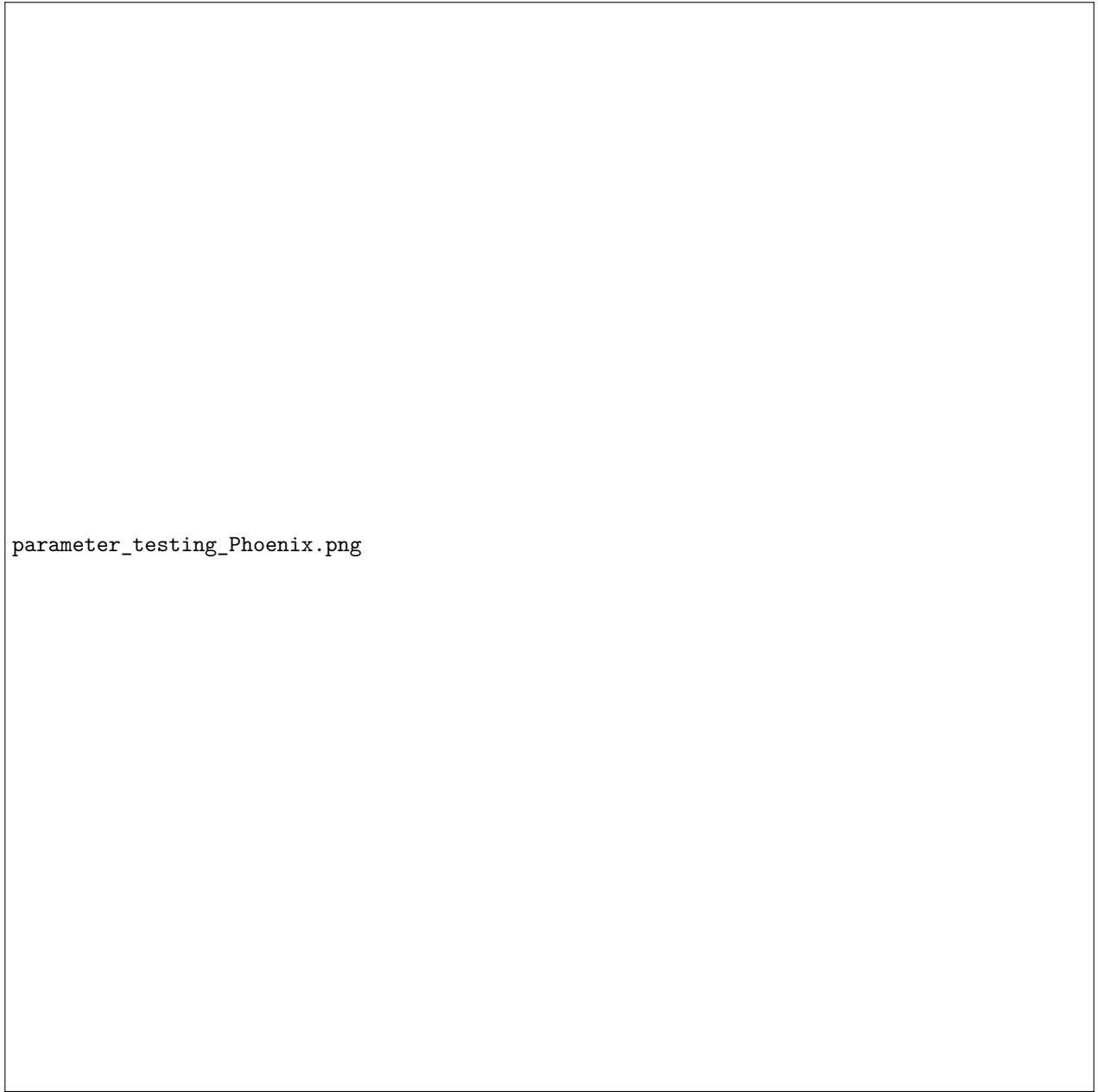
the other for measuring our temperature compliance/violation rate. **NOTE:** Compliance rate is the percentage of trials where the temperature stays within the safe range throughout the day. We also use violation rate, which is defined as the percentage of trials where the temperature goes out of the safe range at least once. The expected daily cost is calculated by taking the average cost across 5000 trials, and the violation rate is calculated by taking the percentage of trials where the temperature goes out of range at least once. We use these metrics to evaluate the performance of our DP policy in all of our tests.

As expected, costs tend to go up and violation rates tend to go down as we increase the penalty parameters. This can be interpreted as the DP policy finding staying within temperature change less costly as the penalty parameters increase, so it is more likely to stay within the safe range by spending more money on power. However, there is a point in Phoenix where increasing the penalty parameters actually slightly increases the violation rate. After the presentation, we understand this is likely due to a bug, since our policy wouldn't be optimal if it's accumulating cost more often for such higher penalty parameters.

One weakness in this experiment is that we are testing very high values of the penalty parameters. Doing this causes the DP policy to be more focused on avoiding temperature violations rather than optimizing cost, which is not the main goal of our project. We are more interested in finding the optimal penalty parameters that balance cost and violation rate, so we should have tested lower values of the penalty parameters as well.

Another interesting result is that setting both of the parameters to 0 results in a nonzero cost. Intuitively, we would expect the cost to be 0 since there is no longer an incentive to stay within the safe range, so the best way to minimize money spent is to spend nothing on power. However, the reason for this cost is that the DP policy has an override feature that when falling out of temperature range. If the temperature falls below $T_{\min}$, the DP policy will turn off the compressor, and if the temperature goes above $T_{\max}$, the DP policy will turn on the compressor at full power. This override feature is still active even when the penalty parameters are set to 0, which is why we see a nonzero cost in this case. That also explains why we get a temperature compliance rate of 0%.

Considering the balance between cost and staying in temperature range, we suggest using a smaller penalty/terminal penalty parameter, such as 10 or 100, since we want the policy to still care about minimizing cost.



parameter_testing_Phoenix.png

Figure 11: Penalty-parameter sweep results for Phoenix.



parameter_testing_Minneapolis.png

Figure 12: Penalty-parameter sweep results for Minneapolis.



parameter_testing_San_Francisco.png

Figure 13: Penalty-parameter sweep results for San Francisco.

Temperature Discretization Step

The current temperature is a significant component of our state space for both scenarios 1 and 2. In an ideal world, we would have a continuous state space, but our implementation decides to discretize instead. This means we have to choose a temperature step size, which is the resolution of our temperature grid. A smaller step size means a finer grid and a more accurate representation of the state space, but would also increase the computational cost of solving the DP problem. A larger step size yields the opposite results, reducing the computational cost but may lead to a less accurate policy.

Our experiments in figure 4, 5, and 6 support our theory. However, there are 2 anomalies in the results. First, all 3 cities have unpredictable behavior in their violation rates as we decrease the temperature step size. We expected the violation rates to decrease as we decrease the temperature step size, since a finer grid should allow for a more accurate policy that can better avoid temperature violations. However, the results show that increasing the step size can sometimes result in a smaller violation rate. Although we are computing violation rates through sampling and could have been unlucky in our draws for smaller step sizes, this is unlikely due to the fact that this behavior is consistent across all 3 cities. We suspect that this is due to a bug in our code, since it doesn't make sense for a coarser grid to outperform a finer grid in terms of violation rates.

The second anomaly is in Minneapolis and San Francisco, where the validated costs exceed the J values. **NOTE:** J values are the costs calculated by our DP algorithm that determine our policy, so they include both energy costs, demand costs,

and the penalties we added for going out of temperature range. The validated costs are the costs we get from running our DP policy on 5000 trials and taking the average cost across those trials, which is composed of only energy/demand costs, not penalty costs. We expected the J values to always be greater than the validated costs, since the J values include the penalty costs while the validated costs do not. However, our Minneapolis and San Francisco contradict this, which we think is due to a bug in our code. One other explanation is our different method of calculating both metrics. J values are calculated by approximating an expectation over the disturbance distribution 200 times, while the validated costs are calculated by taking an average over 5000 samples. So as long as our policy is able to mostly avoid temperature violations, it's theoretically possible but very, very unlikely for the validated costs to exceed the J values due to sampling error, hence why we think the likely explanation is a bug in our code.

Based on our results, we suggest using a temperature step size of 0.1 or 0.05, since these step sizes have reasonably low costs and violation rates, and aren't too computationally expensive to solve.

temp_step_parameter_testing_Phoenix.png

Figure 14: Temperature step sensitivity for Phoenix.



temp_step_parameter_testing_Minneapolis.png

Figure 15: Temperature step sensitivity for Minneapolis.



temp_step_parameter_testing_San_Francisco.png

Figure 16: Temperature step sensitivity for San Francisco.

Power Discretization Step

In the last section, we discussed how the discretization resolution of temperature (temperature step size) affects the performance of our DP policy. In this section, we will test how the discretization resolution of the control input (power) affects the performance of our DP policy. The same anomalies from the previous section are found here, also likely due to the same bugs in our code. Another interesting result is in Phoenix, where the J value jumps dramatically as we increase the power step size from 0.05 to 0.5. We believe this is due to a coarser step size giving the DP policy less control over the temperature, making it more likely to violate the temperature constraints and thus incur penalty costs. Phoenix is an especially hard city to keep within temperature range, due to the high ambient temperature. Since the validated cost does not include penalty costs, it does not change drastically like the J value does.

Based on our results, we suggest using a power step size such as 0.5 and 1.0. This is because a smaller power step size does not seem to significantly improve the performance of our DP policy, while it does increase the computational cost.

power_step_parameter_testing_Phoenix.png

Figure 17: Power step sensitivity for Phoenix.

power_step_parameter_testing_Minneapolis.png

Figure 18: Power step sensitivity for Minneapolis.



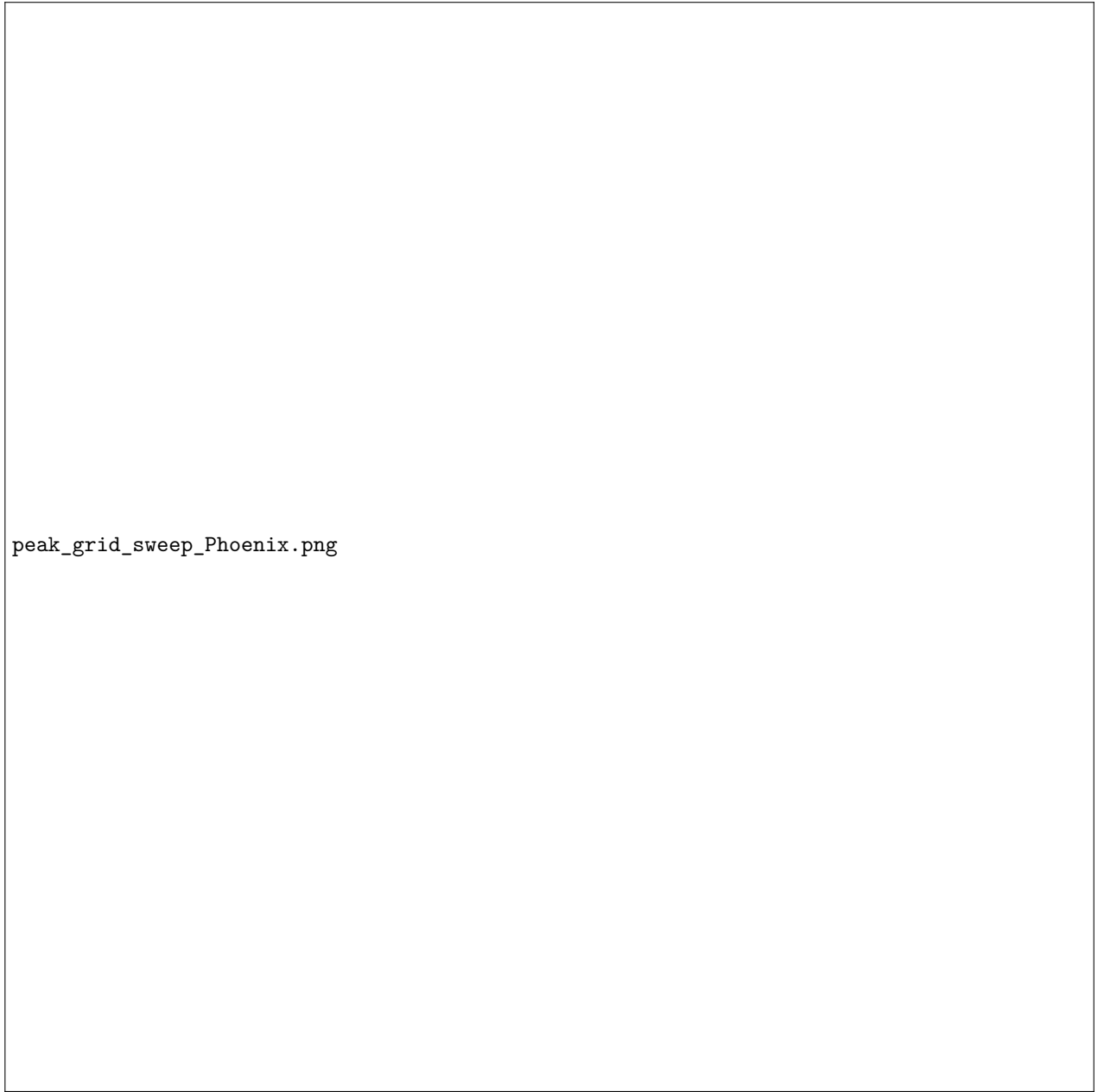
power_step_parameter_testing_San_Francisco.png

Figure 19: Power step sensitivity for San Francisco.

Peak Power Discretization Step

In scenario 2, we added the peak power as a component of our state space, so that we can account for the demand charge in our DP policy. In the last section we tested different power steps, but choosing a small step proved to be computationally expensive. Since demand charge is a significant part of the total cost, we were curious what would happen if we gave the peak power a different discretization level. We tested this by adjusting the peak power step size while keeping the power step size the same.

The results show that adjusting the peak power step size has little effect on the performance at certain point. A step size of 0.5 is adequate for capturing demand cost accurately. However, our results here are likely flawed because the J value for Phoenix actually becomes less expensive as we increase the peak power step, which shouldn't happen. We believe this is due to a bug in our code, since a coarser step size should not lead to a better policy.



peak_grid_sweep_Phoenix.png

Figure 20: Peak power step comparison for Phoenix.



peak_grid_sweep_Minneapolis.png

Figure 21: Peak power step comparison for Minneapolis.

Temperature Padding

Temperature padding is a technique we use to account for when calculating a value in our DP policy requires a temperature that is not on our discretized grid. For example, if we have a temperature range of -25 to -15, and we need to calculate $J(-25)$ and requires $J(-26)$ and $J(-27)$. To do this, we would need to extend our grid beyond just T_{min} to T_{max} . How many more entries to the table is what we try solving in this test. In this experiment, we adjust the number of extra states outside the temperature range to see how performance changes. However, the results show temperature padding has no effect on any metric, besides execution time. At first we thought this was a bug, but then we realized the cause is the policy override feature. When we look outside the temperature range, the policy doesn't bother looking at the J values at these temperatures, but instead decides to either buy max power or shut off the power until we are back within range. This means that the J values outside the temperature range are never actually used, so changing them has no effect on the performance of our DP policy. This was poor design choice on our part, so an improvement we would make given more time is to remove the override feature and instead rely on the J values outside the temperature range to guide the policy back into the safe range.



temp_padding_parameter_testing_Phoenix.png

Figure 22: Temperature padding sensitivity for Phoenix.

temp_padding_parameter_testing_Minneapolis.png

Figure 23: Temperature padding sensitivity for Minneapolis.



temp_padding_parameter_testing_San_Francisco.png

Figure 24: Temperature padding sensitivity for San Francisco.

Expectation Samples

Our DP problem is stochastic, meaning our J values represent the expected minimum cost to finish the day given we start at an initial temperature. This relation is shown in the J value definition for scenario in (19). Calculating the expectation is not trivial due to w being a gaussian distribution and requiring integration, so we instead decide to approximate. Our approximation method is to take a number of samples from the distribution and compute the next temperature for each sample. We can then use our DP table to look up the J values of each of these temperatures, and take the average to get an estimate of the expected J value of our current temperature. This test shows how the number of samples affects the performance of our DP policy. The idea is that the more samples we take, the closer we get to the actual expected J value, which should lead to a better policy.

The test results reveal increasing the number of samples increases our J values and validated costs. This is the opposite behavior we expected, since a better approximation of the expectation should lead to a better policy and thus lower costs. Again, we believe this is a bug in our code.



expectation_samples_parameter_testing_Phoenix.png

Figure 25: Expectation-sample sensitivity for Phoenix.



expectation_samples_parameter_testing_Minneapolis.png

Figure 26: Expectation-sample sensitivity for Minneapolis.



expectation_samples_parameter_testing_San_Francisco.png

Figure 27: Expectation-sample sensitivity for San Francisco.

Monte Carlo Sampling

After we solve our DP problem and get our policy, we need to evaluate how good our policy is. We do this by running hundreds of simulations of the freezer's temperature evolution over a day using our policy and random disturbances drawn from the same gaussian distribution we used in our DP problem. We then calculate the average cost/violation rate across these simulations to get an estimate of the expected performance of our policy. This is called Monte Carlo sampling. However, the number of simulations we run can affect the accuracy of our estimate. If we run too few simulations, our estimate may be inaccurate and not representative of the true expected cost. To test how many samples is a good amount, we try 5 different seeds for our gaussian distribution, and see how the expected cost/violation rate changes as we increase the number of simulations. We expect that as we increase the number of simulations, our estimate of the expected cost will approach the actual cost of our policy and converge.

Our tests confirm our theory. As we increase the number of simulations, the estimates of the expected cost drawn at each seed show trends towards convergence. The same trend happens for the violation rates. Looking at the graphs, we can see that even as little as 500 simulations can give us a good estimate of the expected cost and violation rate, regardless of what seed we decide to use for the gaussian distribution.



monte_carlo_sample_convergence_Phoenix.png

Figure 28: Monte Carlo sample convergence for Phoenix.



monte_carlo_sample_convergence_Minneapolis.png

Figure 29: Monte Carlo sample convergence for Minneapolis.



monte_carlo_sample_convergence_San_Francisco.png

Figure 30: Monte Carlo sample convergence for San Francisco.

Scenario 3: Compressor Selection and Lifecycle Cost

Purpose and Modeling Shift

The first two scenarios asked how best to *run* a given freezer. Scenario 3 asks a different question: which compressor a buyer should install in the first place to run for its lifetime. That shifts what we optimize. The hourly control policy still matters, but it now serves a once-a-year purchasing decision, so the quantity to minimize becomes the annual total cost of ownership:

$$J(q, \ell) = C_{\text{op,yr}}(q, \ell) + \frac{C_{\text{cap}}^{(q)}}{L^{(q)}} + C_{\text{maint}}^{(q)}, \quad (23)$$

Here $q \in \{A, B, C\}$ is the compressor and ℓ the location. Now though, the three quantities now sit on the same cost-analysis: capital spread over the unit's lifetime, annual maintenance, and the operating cost the controller actually drives. We are careful to count only real billed dollars in $C_{\text{op,yr}}$: time-of-use energy plus demand charges. The safety penalties inside the DP shape the policy but never enter the cost we report, like with Scenarios 1 and 2.

Compressor	P_{max} (kW)	β	Capital Cost	Lifetime (yr)	Maintenance/yr
A	8	0.35	\$8,500	8	\$600
B	10	0.50	\$12,000	10	\$450
C	12	0.65	\$16,500	12	\$350

Table 4: Compressor options used for Scenario 3. Compressor C has the highest capital cost, but also the largest cooling authority and best cooling coefficient.

Scoring a full year is where the difficulty begins. Unrolling the Scenario 2 DP over all $24 \times 365 = 8,760$ hours is at once too large to solve at our grid resolution and wasteful, because it throws away structure we already know is present: the price schedule and disturbance statistics repeat every 24 hours, and the demand charge resets every 30 days. The opposite shortcut, solving one representative day and multiplying by 365, fails for another reason. A demand charge is set by the single highest hour in a billing month, so a peak that a daily model quietly amortizes away is, in reality, paid for the whole month. An honest annual cost therefore has to respect the 30-day meter, and the chunked construction in the next two subsections is built precisely to do that.

State, Dynamics, and Compressor-Specific DP

For each compressor-location pair, the DP reuses the Scenario 2 augmented state

$$x_k = (T_k, M_k), \quad (24)$$

where T_k is freezer temperature and M_k is the peak compressor power since the most recent demand-meter reset. For compressor q and location ℓ ,

$$T_{k+1} = T_k + \alpha(T_k^{\text{amb}, \ell} - T_k)\Delta t - \beta^{(q)}P_k\Delta t + \omega_k, \quad M_{k+1} = \max(M_k, P_k), \quad (25)$$

except at a demand-meter reset, where the peak charge is assessed and the next peak state returns to zero. The control grid is $P_k \in \{0, 0.5, \dots, P_{\text{max}}^{(q)}\}$ kW. The temperature grid remains $[-30, -10]$ °C with 0.1 °C spacing, and the disturbance expectation uses 199 antithetic samples per hour. The safety override is retained: if the freezer is too cold, the compressor is forced off; if it is too warm, the compressor is forced to full power. This way, we follow the same logic as with Scenario 2. Note that we did not update the sampling scheme and safety override as will be discussed in the next section on future improvements.

The stage cost used in the Bellman recursion is

$$c_k P_k \Delta t + \lambda_v \phi(T_k), \quad (26)$$

where $\phi(T)$ is the same quadratic violation penalty used in Scenario 2. Demand charges are added only when the meter resets. For a chunk of D days,

$$C_{\text{demand, chunk}} = \begin{cases} \frac{D}{30} c_d M_D, & D \leq 30, \\ c_d \sum_{r=1}^{D/30} M_{30r}, & D > 30, D \in \{60, 120, 180\}. \end{cases} \quad (27)$$

Thus $D = 30$ is the smallest exact physical billing-period model. Shorter chunks are retained only as structural sensitivity checks and longer chunks preserve exact monthly resets while testing whether extra lookahead changes the result.

Chunks, Checkpoints, and Terminal Artifacts

The construction breaks the year into episodic chunks. A chunk is D days long, hence $24D$ hourly stages, and we solve the DP over a single chunk, then annualize by how many chunks fit in a year:

$$C_{\text{op,chunk}} = C_{\text{energy,chunk}} + C_{\text{demand,chunk}}, \quad C_{\text{op,yr}} = \frac{365}{D} \mathbb{E}[C_{\text{op,chunk}}]. \quad (28)$$

Solving one chunk in isolation, though, introduces an artifact. Any finite-horizon DP is distorted near its end by whatever terminal cost we impose. Pin a warm-target penalty at the chunk boundary and the controller over-cools merely to hit it. On the other hand, impose nothing and the final hours turn greedy, dumping any stored cold because, as far as the chunk can tell, there is no tomorrow to pay for it. Neither is real in practice. In continuous operation the next day is just more of the same, temperature carries straight across the boundary, and only the demand meter resets. We restore that continuity with the checkpoint chain in Figure 31: K chunks are solved back-to-back in one backward pass, the artificial terminal is placed only after the very last chunk, and we keep only the *first* chunk’s policy. With enough chunks ahead of it, the first chunk no longer feels the terminal, so what we retain is the policy a continuously-running freezer would actually use.

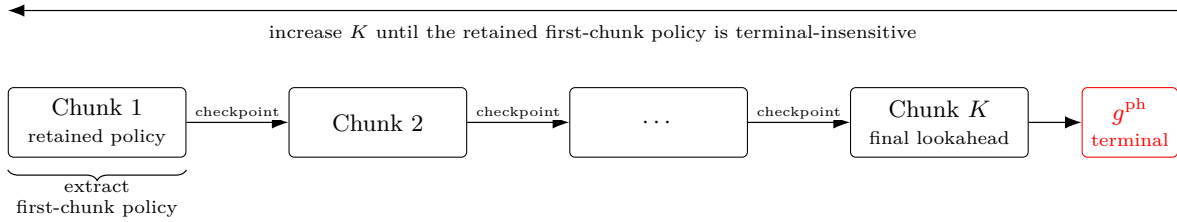


Figure 31: Checkpoint construction. Increasing K pushes the placeholder terminal farther into the future; the retained first-chunk policy should stop changing once the terminal artifact has decayed.

How do we know the chain is long enough? Three sweeps answer that. We vary the placeholder terminal over three choices: zero terminal cost, a safety-only penalty that charges only for ending outside the allowable band $[-25, -15]^\circ\text{C}$, and a warm-target penalty that charges only for ending warmer than $T_0 = -20^\circ\text{C}$. If the construction works, the kept policy should not care which placeholder terminal we used. We grow the checkpoint count K from 1 to 8 and watch the carried ending-temperature distribution, the monthly bill, and the violation hours settle. We also sweep the chunk length over

$$D \in \{1, 7, 14, 30, 60, 120, 180\} \quad (29)$$

to confirm the recommendation is robust to both short-chunk approximations and longer lookahead.

The first thing the sweeps reveal is rather obvious: the placeholder barely matters. Across the three terminals the per-chunk DP objective differs by under 0.08% even at $K = 1$, and by under 0.01% once $K = 8$ (Figure 32). Cost, in other words, is nearly insensitive to the checkpoint count. That is reassuring, but it is not the quantity that decides whether a chunk policy is safe to repeat. The ending *state* is. Under the neutral (zero) terminal a single-chunk solve ends the day warm, with mean carried temperature around -12 to -14°C across the three cities, because nothing in that final chunk discourages coasting up toward the ceiling once the horizon is in view, and a warm hand-off does not repeat cleanly. One additional chunk removes the incentive: from $K = 2$ onward the carried ending temperature settles near -16.4°C in every city and stops moving, which is exactly the stable, repeatable cycle the annualization assumes. The convergence that matters, then, is of the carried state rather than the cost, and it arrives by $K = 2$.



Figure 32: Per-chunk DP objective versus checkpoint count K for the three placeholder terminals (Compressor C at the $D = 30$ chunk). The curves converge as K grows, confirming the retained first-chunk policy becomes insensitive to the terminal choice.

scenario3_k_chunk_state_convergence.png

Figure 33: Carried-state convergence for Compressor C at the physical $D = 30$ chunk. The vertical axis reports annualized operating cost, meaning energy plus demand charge only, while each label reports the mean carried boundary temperature $\bar{T}_D$. The $K = 1$ policy hands off too warm; by $K = 2$ all three cities settle near -16.4°C and additional checkpoint chunks do not change the retained first-chunk behavior.

At the physical $D = 30$ chunk this convergence is reached by $K = 2$ in all three cities (Figure 33). For the production run we nonetheless fix the largest checkpoint count any case in the D -sweep asks for, $K = 8$, since the very short ($D = 1$) and the multi-month ($D \in \{60, 180\}$) chunks settle a little more slowly. We use that value everywhere, together with the zero placeholder and $D = 30$. At $D = 30$ those extra checkpoints leave the kept policy untouched.

Chunk-Length Sensitivity

Does the chunk length itself change the answer? For Phoenix with Compressor C the annualized operating cost eases from \$12,599 at $D = 1$ to \$12,534 at $D = 30$ and is then flat out to $D = 180$. The one-day shortcut is thus close, off by well under a percent, but not exact, whereas $D = 30$ is the shortest horizon that prices the monthly meter with no approximation at all. What matters more for a purchasing decision is that the *ranking* of the compressors does not move anywhere across the whole range (Figure 34): chunk length shifts the third-decimal cost, not which compressor wins.



scenario3_d_ranking_robustness.png

Figure 34: Annualized operating cost versus chunk length D for all three compressors in each location. Each panel lists the compressors that pass or fail the 5% hourly-violation screen, and red crosses mark infeasible points directly on the curves. Compressor A fails in every climate and Compressor B fails in Phoenix; the operating-cost ordering is essentially flat in D , so the compressor recommendation does not change with the chunk-length approximation.

Compressor Ranking at the Physical Billing Period

With the horizon settled at $D = 30$, we can finally rank the three compressors. Every feasible compressor–location case is validated on 5,000 fresh Monte Carlo trials and screened at a 5% hourly-violation threshold; Table 5 collects the annualized result. The “fixed cost” column is capital amortized over the unit’s lifetime plus annual maintenance, so the lifecycle total is directly comparable across compressors that differ in both price and lifespan.

Location	Comp.	Annual Energy Cost	Annual Demand Cost	Fixed Cost Amortized	Lifecycle	Hour viol.	Status
Phoenix	C	\$9,833	\$2,701	\$1,725	\$14,259	0.150%	recommended
Phoenix	A	\$8,468	\$1,801	\$1,663	\$11,931	100.000%	infeasible
Phoenix	B	\$10,585	\$2,251	\$1,650	\$14,486	73.688%	infeasible
San Francisco	C	\$7,003	\$2,420	\$1,725	\$11,149	0.188%	recommended
San Francisco	B	\$9,537	\$2,251	\$1,650	\$13,438	0.298%	feasible
San Francisco	A	\$8,468	\$1,801	\$1,663	\$11,931	100.000%	infeasible
Minneapolis	C	\$6,085	\$2,382	\$1,725	\$10,192	0.196%	recommended
Minneapolis	B	\$8,406	\$2,251	\$1,650	\$12,307	0.248%	feasible
Minneapolis	A	\$8,468	\$1,801	\$1,663	\$11,931	75.123%	infeasible

Table 5: Scenario 3 compressor ranking at $D = 30$ under the 5% hourly violation screen. Dollar values are annualized USD/year and rounded to the nearest dollar.

Finally, after doing the chunk and chain analysis, Compressor C is recommended in all three climates, but for two different reasons. In Phoenix it is the *only* feasible option: the hot-desert load is so high that A and B cannot hold the desired freezing range (A violates essentially every hour, B about 74% of hours). In San Francisco and Minneapolis, Compressor B is also feasible, yet C still wins on lifecycle cost once operating cost is included. Its higher cooling coefficient buys enough energy and demand savings to more than repay its larger capital amortization. Compressor A is the lowest-capital unit and looks inexpensive on paper, but it is disqualified outright by the feasibility gate: with only 8 kW of cooling authority it cannot track the safe band, so it is never a valid choice at any price. In our analysis and project, we have made feasibility a hard constraint that precedes the cost comparison, given the high costs of throwing out spoiled goods in the scenario the freezing range is broken.

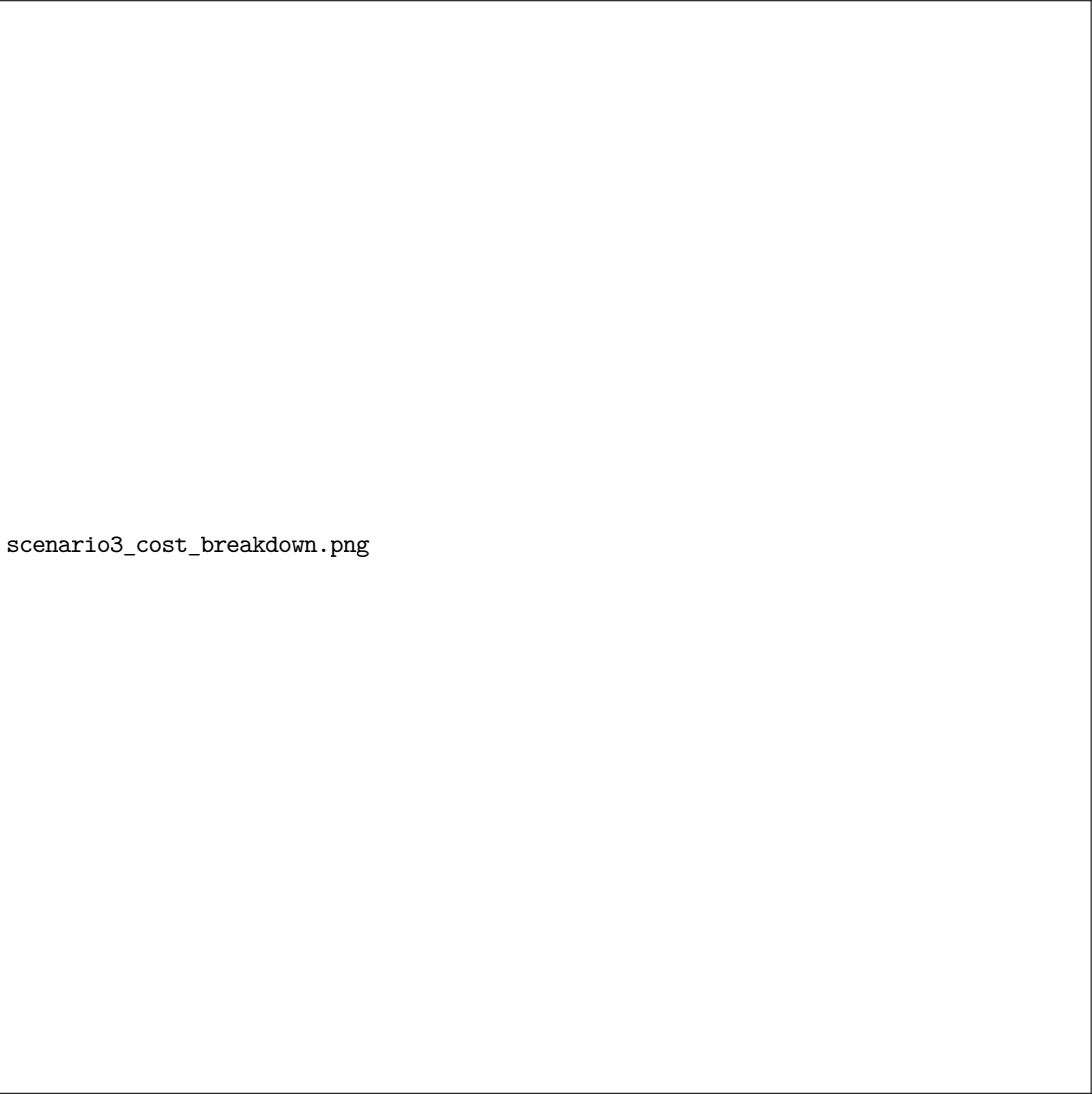
DP Value Against Compressor-C Thermostat Control

Choosing the compressor is equally as important as running it well. To isolate the quality of running the compressor, we hold the cooling hardware fixed at the recommended Compressor C and compare our DP policy against the same full-on/full-off thermostat baseline from Scenario 2, now driving Compressor C. The DP is cheaper to operate in all three cities (Table 6 and Figure 35).

Location	DP-C Operating Cost	Thermostat-C Operating Cost	Annual Savings	DP-C Hour Viol.	Thermostat-C Hour Viol.
Phoenix	\$12,534	\$12,918	3.0%	0.150%	17.368%
San Francisco	\$9,424	\$10,388	9.3%	0.188%	17.821%
Minneapolis	\$8,467	\$9,459	10.5%	0.196%	18.659%

Table 6: Annualized operating-cost and hourly violation-rate comparison for Compressor C at $D = 30$. Annual Savings are computed relative to the full-on/full-off thermostat baseline using Compressor C and exclude amortized capital cost and maintenance.

The major theme of our project has been reliability and staying within the freezing range. The DP with Compressor C has hourly violation rates below 0.2% in all three climates, while the thermostat baseline using Compressor C violates the band roughly 17–19% of hours. This means the DP is using the same compressor more intelligently to preserve feasibility under stochastic disturbances and running a more cost-effective policy.



scenario3_cost_breakdown.png

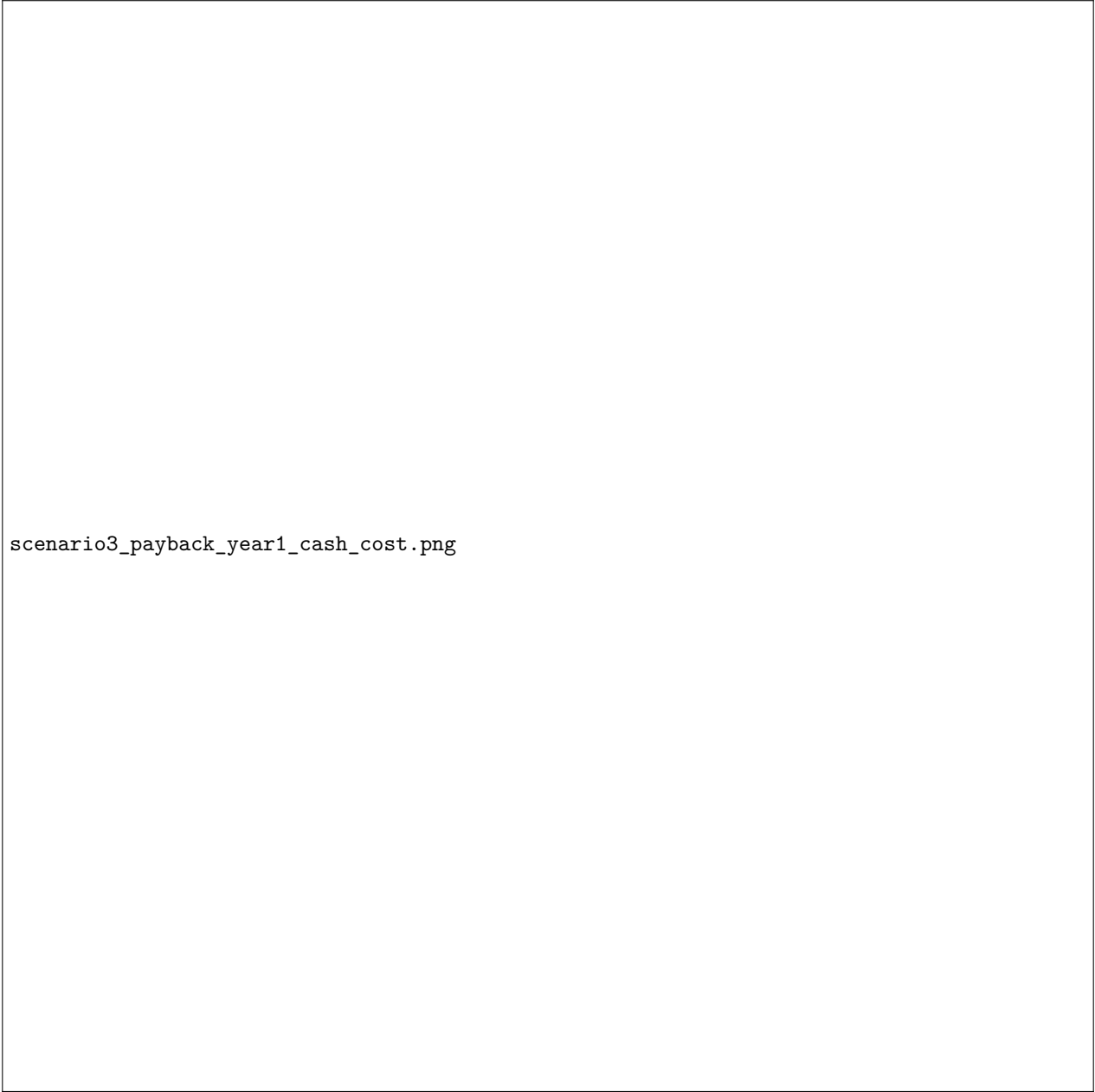
Figure 35: Annualized operating-cost breakdown (energy plus demand) for the DP policy and the full-on/full-off thermostat baseline, both using Compressor C. The DP is lower on both components in every climate. Its reliability advantage, the order-of-magnitude drop in violation hours, is the separate result in Table 6.

Year-One Cash, Payback, and Robustness of the Choice

The lifecycle ranking amortizes each compressor’s capital over its lifetime, which is the right basis for a long-run comparison but hides the heavy capital investment the user faces at the moment. This section reconciles the two views and provides more financial recommendations for a freezer that will run for the lifetime of its compressor.

Feasibility is applied first, as a hard gate: an option that cannot hold the safe band is not a candidate at any price. We therefore drop every option above a 5% hourly-violation rate, namely Compressor A in all three locations, Compressor B in Phoenix, and the full-on/full-off thermostat baseline using Compressor C, and compare only feasible controllers.

In raw first-year cash the ranking inverts. Figure 36 adds upfront capital to the first year of operation and maintenance: where Compressor B is feasible it is the cheaper choice in year one (San Francisco \$24.2k vs. \$26.3k; Minneapolis \$23.1k vs. \$25.3k), because C’s larger capital outlay has not yet been repaid by its lower running cost. Phoenix has only Compressor C after the screen, so no such comparison exists there.



scenario3_payback_year1_cash_cost.png

Figure 36: Year-one cash cost (upfront capital plus first-year operation and maintenance) for the feasible $D = 30$ options. Phoenix retains only Compressor C after the 5% screen; San Francisco and Minneapolis retain Compressors B and C. In year one the higher-capital Compressor C is the more expensive choice wherever B is available.

That gap closes quickly. Incremental ROI is measured against the lowest-capital feasible compressor in each location, since an infeasible cheaper unit is not a real baseline. Upgrading from B to C saves \$2,464/year in San Francisco and \$2,289/year in Minneapolis in operating-plus-maintenance cost, against a \$4,500 capital premium, so the cumulative-cost curves cross within two years: 1.83 years in San Francisco and 1.97 years in Minneapolis (Figure 37). After that, C is cheaper every year. Compressor C is thus not merely the lower-*amortized*-cost option; once the roughly two-year payback is passed it is the better investment on a lifetime, non-discounted, capital-side as well.

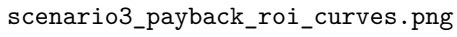
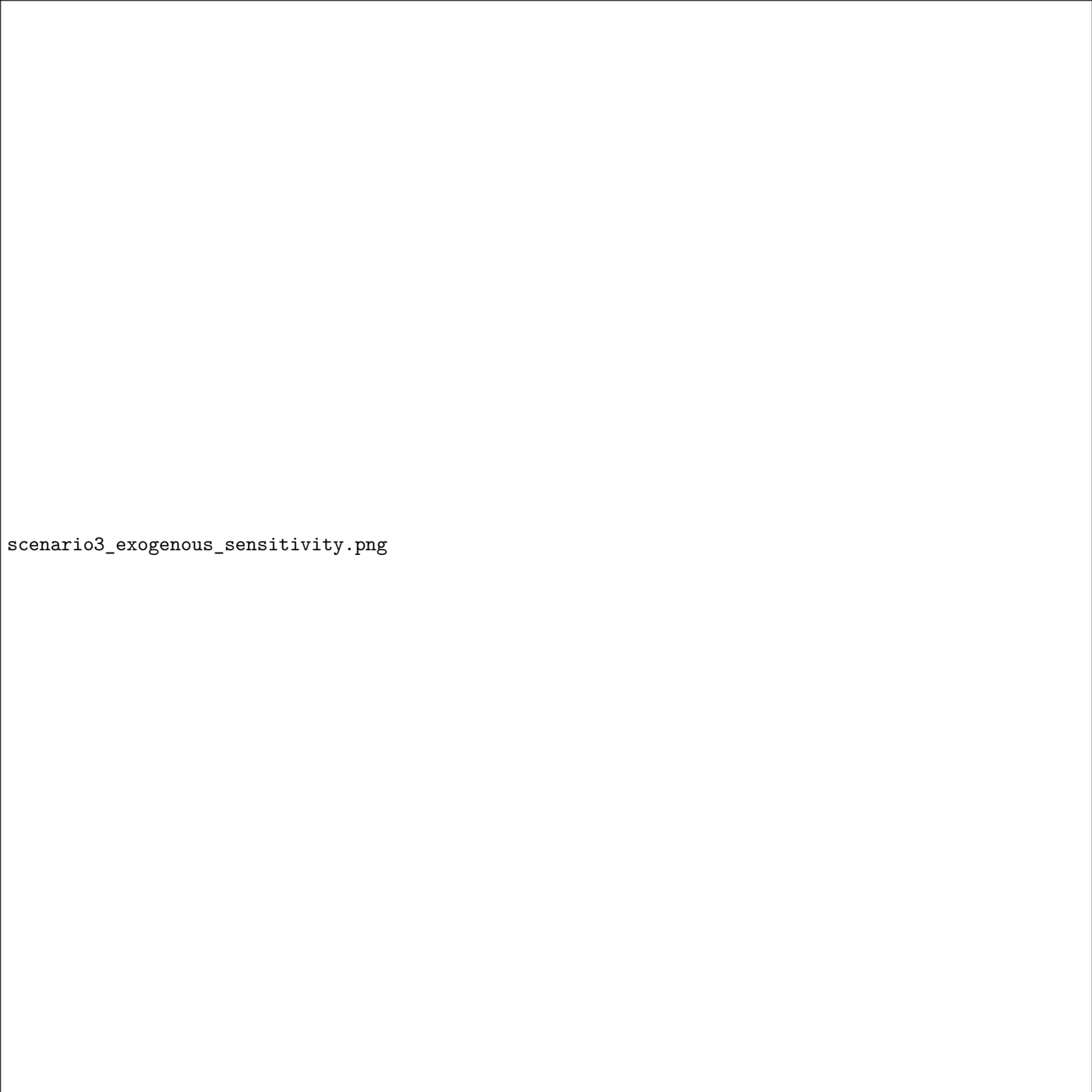
The image is a placeholder for a figure titled 'scenario3_payback_roi_curves.png'. It is represented by a large, empty rectangular box with a thin black border. The text 'scenario3_payback_roi_curves.png' is printed in a monospaced font in the lower-left corner of the box.

Figure 37: Cumulative cash-cost curves for the two locations where Compressor B remains feasible under the 5% screen. Compressor C's higher upfront cost is offset by its lower recurring cost, so the curves cross at roughly two years; beyond the crossover C is cheaper every year.

Finally, the recommendation is re-solved under exogenous shocks rather than merely re-scored. We run 11 shock settings: energy prices are scaled by 0.75, 1.00, 1.25, and 1.50; demand charges are scaled by 0.50, 1.00, 1.50, and 2.00; and disturbance usage is scaled to low, base, and high conditions. For each shock setting the controller is re-evaluated for every compressor-location pair, giving $11 \times 3 \times 3 = 99$ compressor-location evaluations. The actual purchase decision is then made separately in each location by choosing the lowest-cost feasible compressor, giving $11 \times 3 = 33$ location-level recommendation decisions. Compressor C remains the recommended choice in all 33 decisions (Figure 38). The shocks move the cost levels, but they do not change which compressor is selected, so the recommendation is not an artifact of one particular price or load assumption.



scenario3_exogenous_sensitivity.png

Figure 38: Lifecycle cost of all compressor options under exogenous shocks to electricity price, demand charge, and usage intensity. Red crosses mark options that fail the 5% hourly-violation screen; the dotted pale-yellow line marks the recommended feasible option. Compressor C is recommended in every case.

Scenario 3 Takeaways

There are two major takeaways from Scenario 3. First, an honest annual model is not a $365\times$ daily-cost shortcut. It accounts on the 30-day demand meter, carries temperature continuously across chunks, and chains checkpoints so the artificial horizon cannot contaminate the retained policy. We verified that construction converges by $K = 2$ into a stable, repeatable annual cycle. Second, the purchase decision is not simply “buy the most efficient compressor.” Feasibility is a hard gate: Compressor A is the cheapest unit but cannot hold the band. Among feasible options the choice depends on total cost of ownership: energy, demand, annualized capital, and maintenance. On that full score Compressor C is recommended in all three climates; where the cheaper-capital Compressor B is also feasible, C’s roughly two-year payback makes it the better investment for a freezer that is to be run for over 2 years. The recommendation holds under the price,

demand, and usage shock we tested.

Future Work: Scenarios 2 & 3

The problem we found in Scenario 1 (referenced across this report as the “unexplained” bug) was a sampling problem: we estimated an expectation with too few samples drawn from the nominal distribution, and that expectation was dominated by rare tail events (the large out-of-band penalty). So the estimate quietly ignored the unlikely-but-costly cases, the policy came out biased, and we caught it through the (impossible) $P(\text{failure})$ -vs-penalty curve. Switching to importance sampling seemed to fix it.

The same thing occurred in Scenarios 2 and 3, since we again used antithetic sampling to estimate an expected cost-to-go (that the controller minimizes over). Sampling used purely to evaluate a fixed policy afterward is fine, since those draws come from the true distribution and aren’t biased — though any rare-event statistic (such as a small failure rates) still needs enough samples to be trusted.

We haven’t re-run 2 and 3 with importance sampling yet, but the plan is straightforward: find every place an expectation is sampled inside the optimization, swap in importance sampling, and confirm the corrected metric behaves sensibly. We’d also validate the magnitudes with a large brute-force Monte Carlo (much more samples) just to be sure. Conclusions that don’t depend on the tail (cost rankings, cases that fail no matter what) should not change due to importance sampling.

Contributions

- **Scenario 1:** Nitin & Neel
- **Scenario 2:** Chunyu & Zhou
- **Scenario 3:** Mayand
- **Parameter testing:** Ryan
- **Slides and final report:** Everyone completed the slides & report for their contribution listed above